11 A galaxy is approximately 1.2×10^{26} m from the Earth.

(a) Scientists observe light from the distant galaxy.

The wavelength of the observed light is longer than the wavelength of the light emitted from the galaxy.

State the name of this effect.

..... [1]

(b) **(i)** State the current estimate for the Hubble constant H_0.

H_0 = ... [1]

(ii) Calculate the speed at which the galaxy is moving away from the Earth.

speed = ... [2]

(iii) Scientists have measured the speeds at which distant galaxies are moving away from the Earth and their distances from the Earth.

These measurements suggest that all the Universe was once at a single point.

Explain why.

.....

.....

..... [2]

[Total: 6]

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